How Much of the Sun Can We Take Without Turning Off the Lights on Life? A First-Principles Budget for Planetary Solar Harvesting

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Abstract

Earth intercepts about 173,549 terawatts of sunlight — four orders of magnitude more than humanity uses — yet "cover the planet in panels" is the wrong mental model, because most of that light is already doing a job: driving the biosphere. We build a self-contained, first-principles budget that tracks sunlight from the solar constant through the atmosphere (with the ozone layer's share broken out) to the surface, estimates how much energy is deployable on land *without* displacing ecosystems, and compares ground panels with an orbital collector. A conservative desert-and-rooftop footprint already yields 244 TW, about 81× world electricity demand; the entire electricity supply fits on a 274 km square of panel. Space-based solar delivers 6.5× more power per collector square metre, but its advantage is continuous sunlight, not "catching light before the atmosphere." Our headline contribution is an *ecosystem-aware harvesting* problem: each biome must keep a minimum daily light budget, but is flexible about *when*. Casting this as a constrained optimisation, we show the optimum is greedy water-filling — reserve the ecosystem's light during low-value hours — and that this temporal scheduling recovers 17.9% more harvest value than a flat reservation at a 0.3 reserve, rising to 45.4% as the reserve grows. The constraint on solar power is not the Sun; it is the biosphere, and it is schedulable.

1 Introduction

A popular way to frame the renewable-energy question is "how much of the Earth would we need to cover in solar panels?" The framing is seductive because the raw numbers are staggering: the Sun deposits roughly 173,549 TW on Earth's cross-section, against a human primary-energy appetite near 18 TW. By that arithmetic the problem is trivially solved and the only question is cost.

This paper argues that the arithmetic hides the real constraint. The 173,549 TW is not idle: after the atmosphere reflects and absorbs its share, the sunlight that reaches the ground is what powers photosynthesis, evaporation, and every ecosystem on the planet. Harvesting solar energy at civilisational scale is therefore not a question of *area* but of *allocation* — how much light can we divert to electricity before we starve the biosphere that light supports?

We approach this with a deliberately self-contained, first-principles model (no external datasets; every number is reproducible from published physical constants) organised around four questions:

1. **The ceiling.** How much sunlight actually reaches the surface, and where does the rest go? We track the cascade from the solar constant to surface irradiance, breaking out the ozone layer's contribution, and map the resource by latitude band (Section 3).
2. **Feasibility.** How much energy is deployable on land without displacing forests, croplands, or protected habitat, and how does that compare to demand? (Section 4).
3. **The orbital alternative.** Is a collector in space — a "layer" above the atmosphere rather than a Dyson sphere — fundamentally better, and why? (Section 4).
4. **Sharing light with life.** If an ecosystem needs a certain amount of light but is flexible about *when* it receives it, how much more can we harvest by scheduling around it? (Section 5).

The fourth question is our main contribution. We formalise ecosystem-aware harvesting as a constrained optimisation, prove its optimum is a simple water-filling rule, and quantify the energy unlocked purely by respecting *when* — not just how much — light a biome needs.

2 Background and related work

Solar resource physics. The astronomical geometry of insolation — declination, hour angle, and the daily integral of extraterrestrial radiation on a horizontal surface

— is textbook material [2]. The partition of incoming sunlight into reflection, atmospheric absorption, and surface absorption is quantified by global energy-budget studies [8], and the accepted total solar irradiance is 1361 W/m^2 [6]. We reuse these closed forms as computable ground truth rather than fitting data.

Space-based solar power. Collecting sunlight in orbit and beaming it down was proposed by Glaser [4] and has been revisited in modern phased-array concepts [7]. It is the practical cousin of Dyson's stellar-scale collector [3]: rather than enclosing the star, place a collector where the light is undiminished and uninterrupted. We treat it quantitatively as one option among several, not as science fiction.

Sharing land and light with ecosystems. Human appropriation of net primary production already claims a large share of the biosphere's energy throughput [5]. Agrivoltaics shows that photovoltaics and biological light demand can co-exist on the same land when light is shared rather than monopolised [1]. Our ecosystem-aware optimisation abstracts that idea to a schedulable light budget and asks for the allocation that maximises harvest while keeping each ecosystem whole.

Our contribution. We unify the resource ceiling, land feasibility, and the orbital alternative in one reproducible budget, and add a novel optimisation layer: because an ecosystem's light requirement is a daily integral, not an instantaneous one, the harvest schedule has temporal slack we can exploit. We characterise the optimum in closed form and measure the energy it unlocks.

3 The solar resource and its ceiling

From the solar constant to the surface. A disk of Earth's radius intercepts $S_0 \pi R^2 \approx 173{,}549$ TW. Spreading that over the rotating sphere and averaging over day and night divides the peak irradiance S_0 by four, giving a top-of-atmosphere (TOA) horizontal mean of 340 W/m^2 — a value our model reproduces from the daily insolation integral to within 0.5 W/m^2. The atmosphere then takes its share: clouds and aerosols reflect about a quarter back to space, the ozone layer absorbs roughly 2% (almost entirely in the ultraviolet), and water vapour and aerosols absorb a further fifth, leaving a global-mean surface irradiance of 184 W/m^2. Figure 1 (left) shows the full waterfall.

The ozone term. The user's question singled out the ozone layer, and it earns its place in the cascade for a reason disproportionate to its 2% energy share: it removes the biologically destructive UV-B/UV-C band almost entirely. Energetically it is a minor loss; biologically it is the precondition for there being surface ecosystems to share light with at all.

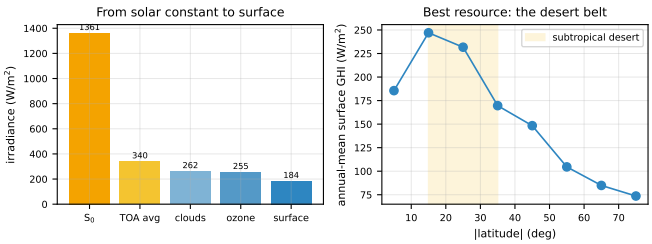


Figure 1: Left: the loss cascade from the solar constant (S_0) through the TOA average, cloud reflection, ozone absorption, and water/aerosol absorption to the surface global horizontal irradiance (GHI). **Right:** annual-mean surface GHI by latitude band — the subtropical desert belt (shaded) is the resource maximum, not the equator.

Where the resource is. Insolation is not uniform. The equator receives the most TOA radiation but sits under a persistent cloud belt; the subtropical desert belt (15–35°) combines high sun angles with the clearest skies and is the true resource maximum, at 247 W/m^2 annual-mean surface irradiance (Figure 1, right). This is why the feasibility analysis that follows places panels in deserts, not on the equator.

4 Feasibility: land, demand, and orbit

How little land it takes. We apply a deliberately conservative "do no ecological harm" filter: eligible land is barren desert (only a 20% usable slice, excluding dunes, reserves, and remote terrain) plus rooftops on built-up land. That leaves 6.1 million km^2. At 20% module efficiency and an 0.8 performance ratio on desert-grade irradiance, this deployable footprint delivers 244 TW on average — about 81× world electricity demand and 13.6× total primary energy (Figure 2). Inverting the calculation, all of world electricity fits on 75,000 km^2 of panel: a single square roughly 274 km on a side. The potential is simply not the bottleneck.

The orbital alternative. If land or intermittency were the constraint, a collector in orbit — a "layer" catching sunlight above the atmosphere — is the natural escape, and a modest cousin of a Dyson sphere. Our chain accounting (Figure 3) puts the delivered power at 261 W per m^2 of collector versus 40 W per m^2 of desert panel, a 6.5× advantage. Crucially, the advantage is *not* mainly from avoiding atmospheric absorption: microwave downlink already passes the atmosphere at ∼95%. It comes from the capacity factor — an orbital collector sees the Sun 99% of the time at full strength, versus an effective 25% on the ground once night and weather are counted. The conversion and transmission chain (PV, DC-to-RF,

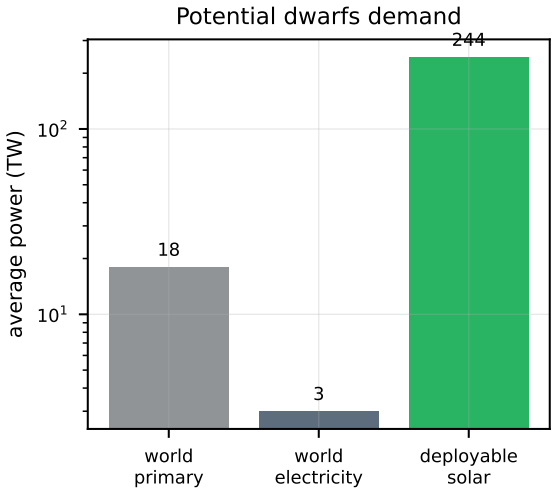


Figure 2: Average power (log scale): world primary energy and electricity demand against the deployable desert-and-rooftop solar resource. Potential exceeds demand by more than an order of magnitude.

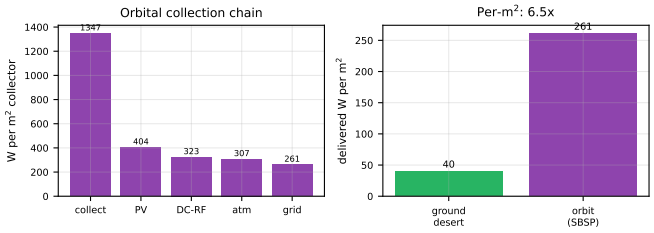


Figure 3: Left: orbital collection-to-grid chain, W per m^2 of collector. **Right:** delivered power per m^2, ground desert panel vs space-based (6.5×).

downlink, rectenna) gives back much of that gain, so orbit is a capacity-factor play, not a free-energy one.

5 Ecosystem-aware harvesting

The previous section shows that raw potential and even land are not the binding constraint. The binding constraint is that surface light is already spoken for by the biosphere. We now make that constraint precise and, crucially, *schedulable*.

Formulation. Partition the surface into cells i (here, latitude bands standing in for biomes) and the day into hours h. Let $L_{i,h} \geq 0$ be the available surface light (from the model of Section 3) and $w_{i,h} > 0$ the *value* of harvesting in that hour — we use a diurnal net-demand ("duck curve") weight, since energy captured at a peak-demand hour is worth more than energy captured at midday when supply is already abundant. We choose a harvest fraction $a_{i,h} \in [0, 1]$; the ecosystem receives the reserved light $(1 - a_{i,h})L_{i,h}$. Each cell must keep a daily light integral $D_i = f_{\min} \sum_h L_{i,h}$ to remain intact:

$$\max_{a \in [0,1]} \sum_{i,h} w_{i,h}\, a_{i,h}\, L_{i,h} \quad \text{s.t.} \quad \sum_h (1 - a_{i,h}) L_{i,h} \geq D_i \;\; \forall i. \tag{1}$$

The optimum is water-filling. Reserving one unit of light in hour h forfeits $w_{i,h}$ of harvest value, so to meet the quota D_i at minimum cost we reserve the ecosystem's light from the *lowest-value* hours first, up to each hour's availability. This greedy rule is the exact optimum of the per-cell linear program in (1); we verify it against `scipy`'s LP solver to 10^{-6} on random instances. Intuitively: give the biosphere its light when we least want to sell it, and harvest the peaks.

Baselines. We compare three feasible-or-bounding policies: (i) *no-cap*, harvest everything ($a \equiv 1$) — an ecosystem-blind upper bound that would collapse the biome; (ii) *flat*, reserve the same fraction $f_{\min}$ every hour, which is feasible but wastes valuable peak-hour light; and (iii) *optimal*, the water-filling schedule. The flat and optimal policies reserve *exactly the same* total light — the ecosystem is equally protected — so any difference between them is pure scheduling gain.

6 Results

Scheduling around life pays. At a headline reserve of $f_{\min} = 0.3$ — the ecosystem keeps 30% of its daily light — the flat policy retains 70% of the ecosystem-blind harvest value, while the optimal temporal schedule retains 82.6%. That is a 17.9% gain in harvested value at *identical* ecological cost, obtained purely by choosing *when* to yield light to the biome (Figure 4). The gain grows with the reserve: the more light the ecosystem needs, the more a smart schedule matters, reaching 45.4% at $f_{\min} = 0.6$.

What the schedule looks like. Figure 5 shows the optimal harvest fraction across bands and hours at $f_{\min} = 0.3$, overlaid with the demand curve. The schedule yields light to the ecosystem around midday — when supply is abundant and marginal harvest value is low — and harvests aggressively during the morning and evening demand peaks. This is the concrete answer to the motivating question: an ecosystem that needs, say, 30% of its daily light can be given that light in the hours we least want it, leaving the valuable hours for harvesting.

Putting the numbers together. The picture is consistent across all four questions. The Sun offers 173,549 TW; the surface receives 184 W/m^2 on average; a conservative desert-and-rooftop footprint already exceeds electricity demand 81-fold; orbit multiplies per-collector power 6.5× but chiefly by buying capacity factor;

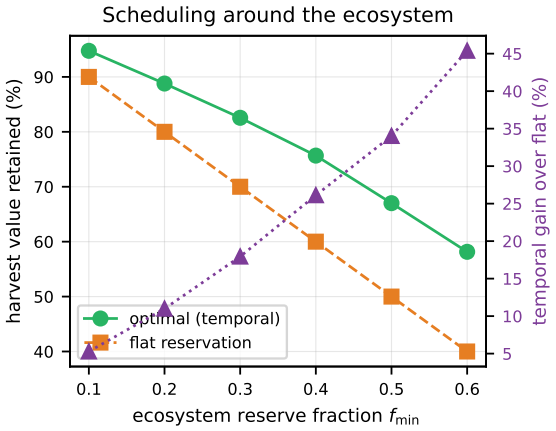


Figure 4: Harvest value retained vs the ecosystem reserve fraction $f_{\min}$, for the optimal temporal schedule and the flat reservation (both protect the ecosystem equally). The temporal gain (right axis) widens as the reserve grows.

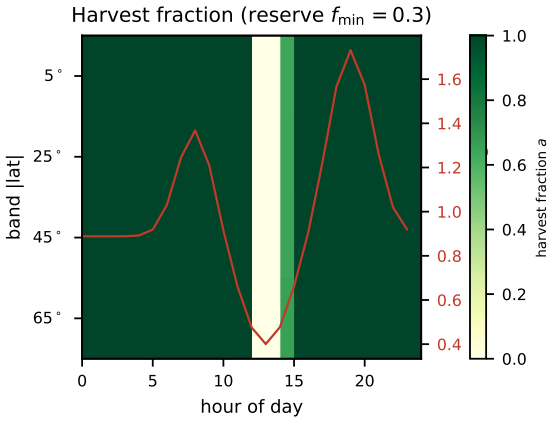


Figure 5: Optimal harvest fraction a (green = harvest, pale = reserved for the ecosystem) by latitude band and hour at $f_{\min} = 0.3$, with the demand weight (red). Light is yielded to the biosphere at low-value midday hours.

and once we accept that surface light belongs to the biosphere, the way to harvest more is not more area but better timing — worth up to 45.4% here. The bottleneck was never the Sun.

7 Discussion and conclusion

We set out to answer a simple-sounding question — how much of the Sun's energy can humanity take — and found that the honest answer inverts the usual framing. The resource is astronomically abundant (173,549 TW), land is not scarce (world electricity fits on a 274 km square of desert panel), and even orbit, the "layer above the atmosphere," is a capacity-factor improvement rather than a new source of free power. The one real, physical constraint is that the light reaching the ground is already running the biosphere.

The contribution we are most confident in is turning that constraint into an optimisation. Because an ecosystem's light requirement is a daily *integral*, harvesting has temporal slack, and exploiting it — yielding light to life at low-value hours and harvesting the peaks — recovers 17.9% more usable energy at $f_{\min} = 0.3$ and up to 45.4% at higher reserves, all at zero additional ecological cost. The optimum is a transparent water-filling rule, not a black box.

Limitations. The model is deliberately coarse and self-contained: latitude bands stand in for biomes, the ecosystem constraint is a single daily-integral floor rather than a species- and process-specific light response, and demand is a stylised duck curve. The `gpu_full` config refines the spatial and temporal grids and the reserve sweep; substituting gridded insolation and land-cover data, per-biome light-response curves, and real regional demand would sharpen the numbers without changing the structure. The qualitative claim — that solar harvesting is an allocation problem against the biosphere, and that the allocation is schedulable — is what the model is built to show.

Outlook. The framing suggests a research programme: replace the integral floor with measured net-primary-production light responses, couple the harvest schedule to storage and grid dynamics, and let space-based collectors relax the temporal constraint entirely for the hours the biosphere claims. The Sun is not the limit; the biosphere is, and it is one we can plan around.

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